

Predicting Top-Tier Universities from World-Ranking Indicators

TEAM 05

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Abstract. This study develops binary classifiers to identify Top-tier universities using indicators from the Times Higher Education, Academic Ranking of World Universities (Shanghai), and CWUR ranking systems. A university is labelled `top_tier = Yes` when its average world rank across the three sources is less than or equal to 100. After integrating and preprocessing 555 university-year observations, models were evaluated on the same stratified held-out test set. Full-feature Logistic Regression achieved the strongest overall performance: accuracy 0.982, precision 0.970, recall 0.985, F1-score 0.977, Cohen's kappa 0.962, and ROC-AUC 0.999. It was therefore selected over a Random Forest comparator. The results show that the available ranking indicators effectively reconstruct the defined Top-tier threshold; however, because the target and predictors originate from the same ranking systems, they do not demonstrate independent forecasting of future university quality.

1. Introduction and research questions

University rankings combine information about teaching, research, citations, internationalisation, and institutional characteristics. This project investigates whether the performance indicators available in three ranking datasets can distinguish universities that are Top-tier under a transparent operational definition.

The task is binary classification. A university is classified as Top-tier (`Yes`) when the average of its Times, Shanghai, and CWUR world ranks is less than or equal to 100; all remaining universities are classified as `No`. This consensus-based operational definition is used consistently throughout the preprocessing, training, and evaluation phases. The study addresses three questions:

1. Which available metrics distinguish Top-tier universities from the others?
2. How predictive are the available ranking indicators for this classification task?
3. Do linear and nonlinear models obtain different results?

The supplied KNIME workflow is the authoritative implementation of the data processing, training, and evaluation steps.

2. Dataset and preprocessing

The World University Rankings dataset contains the Times, Shanghai, and CWUR source tables. The `school_and_country_table` was also loaded as a possible source of country-level enrichment, but it was not used in the final modelling table. University names were conservatively normalised by converting them to lowercase and trimming leading and trailing spaces. The three ranking-source tables were then integrated through inner joins on `university_name + year`. The resulting integrated table contains 555 university-year observations for institutions present in all three sources in the corresponding year.

World-rank ranges such as 100–210 were converted to their lower bound and numeric form. `average_world_rank` was calculated from the three world ranks and used only to derive the target. Textual numeric fields were converted to numbers; the female/male ratio was split into `times_female_value` and `times_male_value`. Columns with substantial structural missingness

(`times_total_score` and `shang_total_score`) were removed; the remaining numeric missing values were median-imputed.

Table 1. Preprocessing and modelling-table summary.

Aspect	Decision	KNIME implementation
Sources	Times, Shanghai, and CWUR; country table loaded but not used	Four CSV Reader nodes
Integration	Inner joins on normalised <code>university_name</code> + <code>year</code>	String Manipulation; Joiner
Final observations	555 university-year observations	Statistics validation
Target	<code>top_tier</code> : Yes if <code>average_world_rank</code> $\leq$ 100	Math Formula; Rule Engine
Numeric conversions	Ranks, percentages, student counts, and ratio components	String Manipulation; String To Number
Missing values	Structural-score columns removed; remaining values median-imputed	Column Filter; Missing Value
Predictors	26 numeric variables: 10 Times, 6 Shanghai, 9 CWUR, and <code>year</code>	Final Column Filter
Leakage prevention	World ranks and <code>average_world_rank</code> excluded from inputs	Final Column Filter

Identifiers and world-ranking variables were excluded from inputs because the ranks directly define the target. National-rank fields were excluded to maintain a homogeneous predictor schema.

3. Experimental design and KNIME workflow

The data were divided with KNIME's `Partitioning` node into 70% training and 30% test data, using stratified sampling on `top_tier` and random seed 42. The same fixed stratified hold-out split, containing 167 test observations, was reused for every model. This ensures a direct comparison between learners, although it does not measure performance variability across different splits.

The workflow preprocesses the sources, filters identifiers and leakage variables, then partitions the modelling table. Training data feed the baseline and Logistic Regression branch and the Random Forest branch; the test data feed Predictor, Scorer, and ROC Curve nodes.

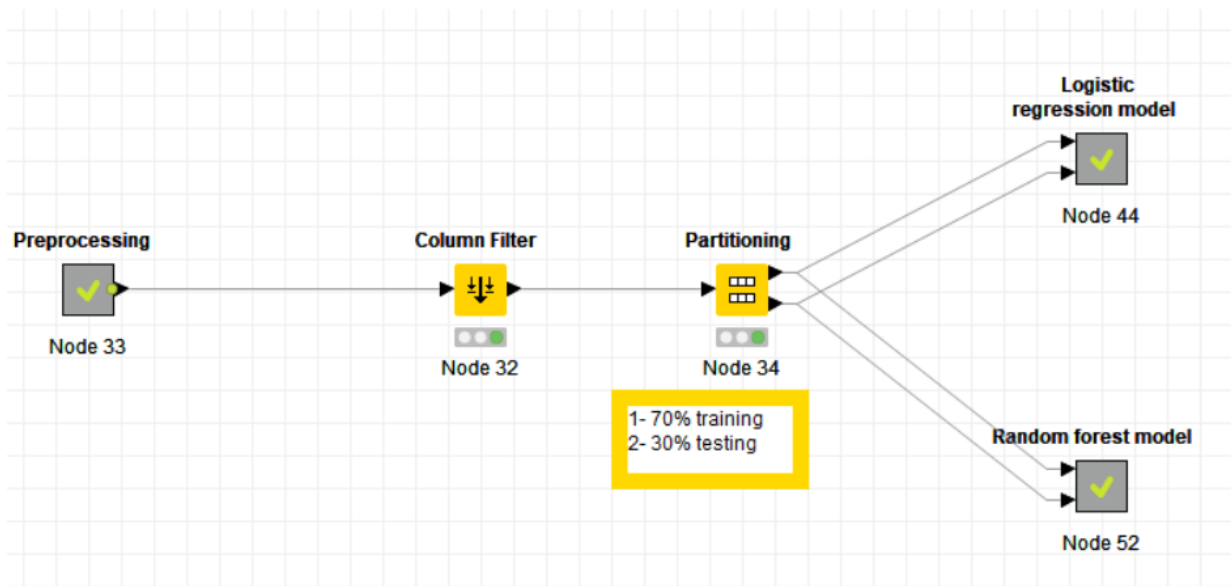


Figure 1. Main modelling branch of the KNIME workflow, from preprocessing and leakage control to the shared train/test partition and the Logistic Regression and Random Forest learners.

A dummy majority-class baseline was included. Logistic Regression used all 26 features, No as reference, an intercept, and default settings. The nonlinear comparison used a Random Forest configured with the Gini index, 100 trees, a maximum depth of 5, and seed 42.

4. Results and model comparison

The majority-class baseline predicted every test observation as No. It achieved accuracy 0.611 but did not identify any Top-tier university. Both learned models substantially improved on this reference. The stratified test set contained 65 Top-tier and 102 non-Top-tier observations.

Table 2. Held-out test-set comparison (167 observations).

Model	Accuracy	Precision (Yes)	Recall (Yes)	F1 (Yes)	Kappa	ROC-AUC
Dummy majority baseline	0.611	0.000	0.000	0.000	0.000	—
Logistic Regression, full features	0.982	0.970	0.985	0.977	0.962	0.999
Random Forest, 100 trees, depth 5	0.976	0.955	0.985	0.970	0.950	0.997

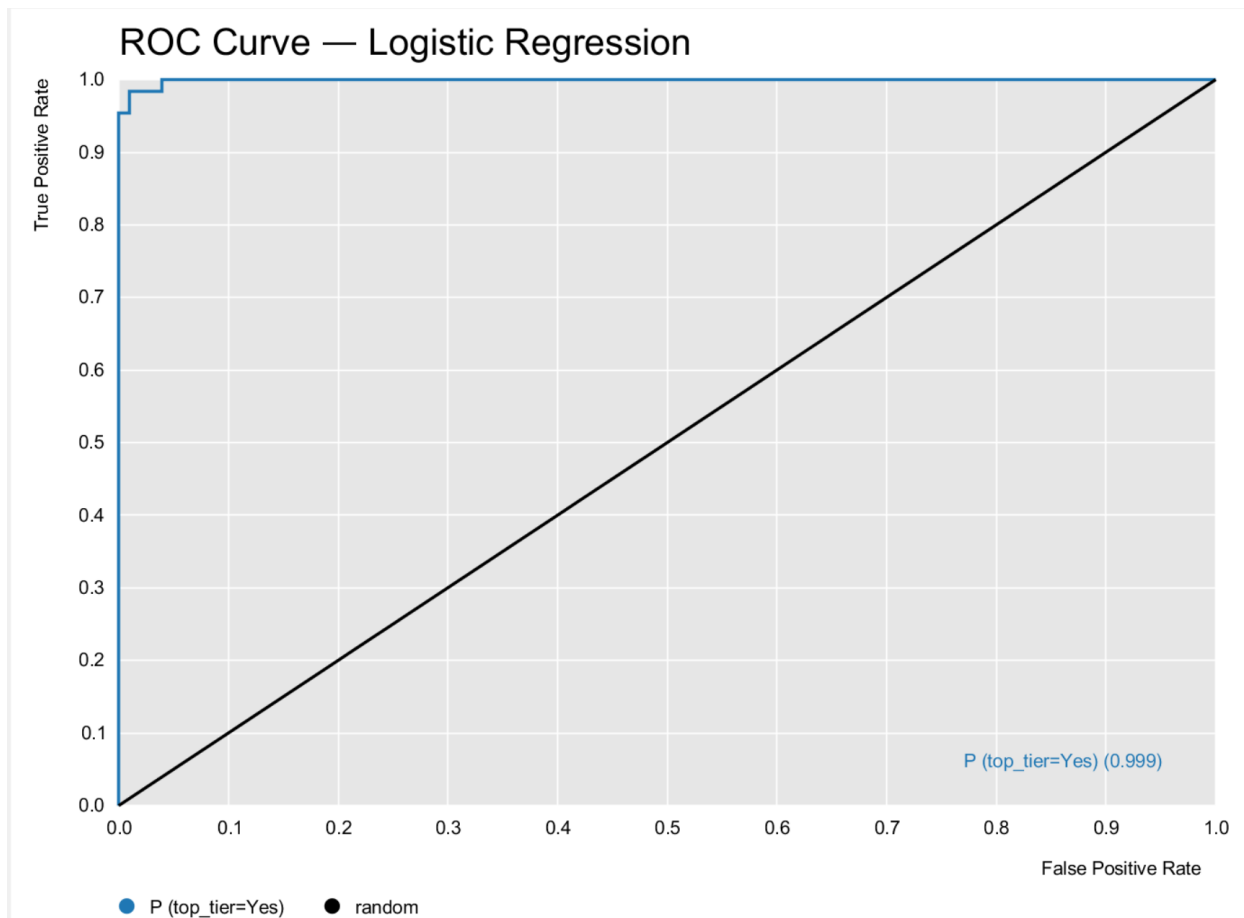


Figure 2. ROC curve for the selected Logistic Regression model on the fixed held-out test set. The model achieved ROC-AUC = 0.999; the black diagonal represents random classification.

Logistic Regression correctly identified 64 of 65 Top-tier universities and produced two false positives. Random Forest reached the same positive-class recall but produced one additional false positive. A light Random Forest tuning experiment did not improve the results. Removing `times_male_value` preserved accuracy, F1-score, and kappa, but reduced recall from 0.985 to 0.969; therefore the full-feature model was retained.

Table 3. Confusion matrix for the selected full-feature Logistic Regression.

Actual class	Predicted Yes	Predicted No
Yes	64	1
No	2	100

5. Discussion, limitations, and conclusion

The indicator set clearly distinguishes the two operational classes: the selected model achieves 0.985 recall and 0.977 F1-score for Top-tier universities. The present analysis does not identify a reliable ranking of individual metrics. The Logistic Regression coefficient table showed extremely large standard

errors, near-zero z-scores, and p-values of one, consistent with strong multicollinearity, different feature scales, and possible quasi-separation. Individual coefficient signs and p-values should not be interpreted as evidence that specific metrics are independently significant.

The indicators are highly predictive for the defined `top_tier` task, but the target is constructed from ranking variables while predictors are other measures from the same systems. The model therefore reconstructs a closely aligned threshold; it is not an external or future forecast of university quality. Logistic Regression is slightly better than Random Forest on every reported metric except recall, where both achieve 0.985, and is simpler to reproduce and communicate.

The main limitations are the single fixed split and shared origin of target and predictors. Median imputation occurred before partitioning, so its statistics may incorporate information from the complete table; a stricter design would estimate medians on training data only. Future work could use cross-validation and standardised collinearity-focused analysis. In conclusion, full-feature Logistic Regression is the preferred model for the operational Top-tier threshold.

References

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